

DETROIT WAYNE COUNTY AP, MI, USA

WMO: 725370

Lat: 42.231N

Lon: 83.331W

Elev: 192

StdP: 99.04

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.7	-21.6	0.6	-16.2	-18.7	0.7	-12.8	12.6	-2.4	11.4	-1.3	4.1	250	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.4	23.3	30.8	22.5	29.4	21.6	24.8	30.1	23.9	28.6	23.0	27.5	5.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.4	27.7	22.3	17.4	26.7	21.4	16.4	25.8	76.2	30.2	72.3	28.7	68.8	27.6	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.4	8.4	-19.8	35.0	3.8	1.5	-22.5	36.1	-24.7	36.9	-26.9	37.8	-29.6	38.8		
(5)			-20.2	26.6	3.7	1.1	-22.9	27.4	-25.1	28.0	-27.1	28.7	-29.8	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-3.6	-2.2	2.9	9.5	15.7	21.2	23.4	22.5	18.6	11.9	5.2	-0.4		
	DBStd	10.61	6.14	5.68	5.81	4.93	4.64	3.68	3.01	2.80	3.99	4.64	4.93	5.20		
	HDD10.0	1582	422	342	232	67	6	0	0	0	1	32	158	323		
	HDD18.3	3353	680	574	479	268	109	14	1	3	44	207	393	580		
	CDD10.0	1751	1	1	12	51	183	336	414	388	259	90	14	2		
	CDD18.3	482	0	0	1	3	28	101	157	132	54	7	0	0		
	CDH23.3	3871	0	0	6	37	259	830	1358	964	371	45	1	0		
	CDH26.7	1187	0	0	1	6	68	277	461	269	99	7	0	0		
(14) Wind	WSAvg	4.0	4.7	4.5	4.4	4.5	3.9	3.5	3.3	3.1	3.2	3.9	4.3	4.4		
(15) Precipitation	PrecAvg	828	55	52	56	77	89	83	82	81	75	67	59	53		
	PrecMax	1094	103	118	120	135	198	148	151	149	148	176	146	116		
	PrecMin	612	16	18	15	16	29	29	35	8	20	6	14	15		
	PrecStd	111	23	23	28	28	43	33	30	39	40	33	29	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	16.2	23.3	27.4	30.8	34.0	34.7	33.4	32.2	27.5	20.2	15.3		
		MCWB	11.9	12.0	16.1	18.3	21.1	23.6	24.9	24.0	22.9	20.0	13.9	13.1		
	2%	DB	10.6	11.4	18.5	23.8	28.6	31.7	32.6	31.1	29.5	24.3	17.1	12.0		
		MCWB	9.1	8.5	12.7	15.4	19.8	22.5	23.6	23.0	21.4	18.1	13.2	10.0		
	5%	DB	7.3	7.9	14.9	20.9	26.5	29.8	30.9	29.5	27.2	21.8	15.1	9.1		
		MCWB	5.5	5.3	10.8	13.6	19.1	21.5	22.6	22.1	20.2	16.6	11.9	6.9		
	10%	DB	4.3	5.0	11.7	18.0	23.9	28.0	29.3	28.0	25.3	19.5	12.7	6.4		
		MCWB	2.8	2.9	7.9	12.0	17.5	20.6	21.7	21.3	19.3	15.1	9.7	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.9	16.8	19.1	22.8	25.2	26.6	25.8	24.3	21.5	16.1	13.4		
		MCDB	13.5	14.7	22.7	25.7	28.9	32.1	32.4	31.4	29.8	26.1	18.2	15.0		
	2%	WB	9.4	8.9	13.8	16.9	21.3	23.8	25.1	24.5	22.9	19.5	14.1	10.5		
		MCDB	10.4	11.2	17.1	21.8	26.5	29.6	30.5	29.0	27.3	22.7	16.4	11.8		
	5%	WB	5.6	5.3	11.3	14.7	20.0	22.8	24.0	23.5	21.6	17.5	12.0	7.4		
		MCDB	7.0	7.6	14.6	19.6	24.9	28.2	28.6	27.5	25.3	20.9	14.5	9.1		
	10%	WB	2.9	3.0	8.0	12.7	18.4	21.8	23.0	22.5	20.4	15.7	10.0	4.6		
		MCDB	4.1	4.8	11.3	17.3	23.0	26.3	27.4	26.3	24.0	18.8	12.5	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.6	9.1	10.4	10.5	10.1	10.1	9.5	10.0	9.3	7.7	6.2		
		MCDBR	9.4	11.2	13.8	14.7	12.9	11.9	11.2	10.6	11.9	12.2	10.9	8.9		
	5% WB	MCWBR	8.5	9.0	9.3	8.8	6.6	5.5	5.1	4.9	5.5	7.0	8.0	7.7		
		MCWBR	9.2	10.7	13.0	13.2	11.5	10.4	10.0	9.3	10.3	10.7	9.9	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.338	0.355	0.402	0.436	0.452	0.443	0.436	0.392	0.361	0.319	0.313		
	taud		2.395	2.273	2.342	2.232	2.188	2.183	2.233	2.242	2.343	2.410	2.493	2.400		
	Ebn at Noon		840	855	888	866	843	827	831	824	838	823	818	789		
	Edh at Noon		79	105	111	134	144	145	137	131	110	92	73	73		
(44) All-Sky Solar Radiation	RadAvg		1.52	2.40	3.51	4.61	5.39	6.06	6.09	5.28	4.27	2.79	1.74	1.22		
	RadStd		0.14	0.30	0.35	0.40	0.47	0.39	0.31	0.20	0.37	0.24	0.23	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	N/A

Nomenclature: See separate page